



AEGIS

Anonymized Exchange Gateway
for Imaging Studies

Brain MRI PET CT DICOM

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A GCP-native platform for secure, HIPAA-compliant sharing of medical imaging data across all DICOM modalities

The Problem

Hospitals and research institutions need to share medical imaging data — MRI, CT, PET, ultrasound, X-ray, mammography, nuclear medicine, and more — for multi-site clinical trials,

research collaborations, and second opinions. Today this requires:

- **Installing desktop software** (Java apps, Electron clients, CLI tools) at every sending site
- **Manual de-identification** prone to human error, leaving PHI exposed
- **No automated defacing** — facial features in 3D head scans enable re-identification
- **Self-hosted DICOM servers** (XNAT, Orthanc) with high maintenance burden
- **Fragmented compliance** — each site manages its own HIPAA controls
- **Modality limitations** — many platforms are built for a single specialty (neuroimaging, cardiology) and don't generalize

Result: Data sharing is slow, risky, and expensive. Studies are delayed by months.

The Solution

AEGIS is a cloud-hosted platform that makes secure medical image sharing as easy as uploading a file — for **any DICOM modality**.

Capability	How It Works
Zero-install anonymization	Browser-based DICOM tag de-identification. No software to install at hospital sites. Works with any DICOM data.
Modality-agnostic	Supports all DICOM modalities: MRI, CT, PET, PET/CT, US, X-ray, mammography, NM, SPECT, RT, and more.
Automated defacing	Server-side facial feature removal from head scans. Non-head studies bypass defacing automatically.
HIPAA by design	Client-side PHI stripping before data leaves the hospital network. VPC-SC, CMEK, audit trails.
Managed DICOM storage	GCP Healthcare API — DICOMweb endpoints, no servers to maintain.
AI-powered QC	Vertex AI detects burned-in PHI, assesses image quality, flags issues automatically.

OHIF web viewer for QC, defacing review, and routing — all in the browser.

AEGIS Architecture
Anonymized Exchange Gateway for Imaging Studies

SENDING SITE (Hospital Browser — Zero Install)

- React Upload Portal (PWA)**
 - DICOM fetcher/picker
 - Client-side parsing (dcmjs)
 - Tag de-identification (P53.15)
 - Anonymization preview
 - Chunked upload to GCS
- De-identification Engine**
 - HIPAA Safe Harbor (18 identifiers)
 - D32X/LUC action codes per tag
 - Deterministic UID hashing
 - Date shifting
 - Private tag removal
- Uploader Library (TypeScript)**
 - Reusable npm package
 - Web Workers for parsing
 - Embeddable in other apps
 - Progress tracking
- Key Principle**
PHI is stripped in the browser BEFORE data leaves the hospital network.
Only tag-de-identified DICOM is transmitted over HTTPS/TLS.
No software installation required.

HTTPS (TLS 1.2+) — De-identified DICOM only

GCP PROJECT (Secured Enterprise Tenancy)

- Cloud Armor (DDoS / WAF)**
 - Global HTTPS Load Balancer
 - Identity-Aware Proxy (admin routes)
- API Backend (Go / Cloud Run)**
 - Upload orchestration (signed URLs)
 - Study / project management
 - User auth (OAuth 2.0 / JWT)
 - Routing rules engine
 - DICOMweb proxy @ Healthcare API
 - Pub/Sub event handlers
 - distroless image (~10-20 MB)
- GCP Healthcare API**
 - DICOM Store (DICOMweb)
 - STOWRS / WADO-RS / DICOMweb
 - Server-side de-identification
 - DICOM type detection (DLP)
 - Pub/Sub notifications
 - BigQuery metadata export
 - CMEK encryption at rest
- Cloud Storage**
 - Staging bucket (uploads)
 - Archive bucket (defaced)
 - Signed URL uploads
 - Lifecycle policies
- Pub/Sub**
 - Ingest notifications
 - Defacing triggers
 - Routing events
 - Audit events
- Defacing Service (Python / Cloud Run)**
 - Triggered for heat MRS/PET/CT
 - WADO-RS retrieval from DICOM store
 - dcm2nix (DICOM @ NIFTI)
 - mri_deface or DeepDefacer
 - pydicom pixel data injection
 - STOW-RS back to 'clean' store
 - ~0.5-2 GB container image
 - 2-10 min per volume
- Admin Dashboard (React / Cloud Run)**
 - Behind identity-aware Proxy (IAP)
 - OHIF Viewer (DICOMweb)
 - Study browser + QC review
 - Defacing review (before/after)
 - Routing rule configuration
 - User/institution management
 - Audit log viewer
- Two DICOM Stores**
 - raw**: Tag de-ids, not defaced
 - clean**: Fully processed + defaced
 - Non-destructive pipeline: raw D → deface D → clean D → approve
 - Admin reviews before release
- Security Layers**
 - VPC Service Controls
 - Cloud Armor DDoS/WAF
 - TLS 1.2+ everywhere
 - CMEK (Cloud KMS)
 - IAM least privilege
 - Cloud Audit Logs
 - Secret Manager
 - Artifact Registry scanning
 - distroless containers

Repository Structure (5 Repos)

- aegis-terraform-prj**: GCP project bootstrap, IAM, KMS
- aegis-terraform-infra**: VPC, Cloud Run, Healthcare API, Cloud Armor
- aegis-api**: Go backend — upload, routing, DICOMweb proxy
- aegis-frontend**: React — Upload Portal + Admin Dashboard
- aegis-client**: TypeScript uploader library (npm package)

Key Open-Source Dependencies

- Go**: sayashkumar/dicom, GCP Go SDK
- Browsers**: dcmjs, dicomParser, OHIF Viewer
- Defacing**: mri_deface, dcm2nix, pydicom, nibabel
- Infra**: Terraform Google Provider, Cloud Build
- Viewing**: OHIF Viewer (MIT) — DICOMweb native

Implementation Phases

- Phase 1: Foundation MVP**
Terraform + Go API + Upload Portal + Admin Dashboard
- Phase 2: Defacing Pipeline**
Python Cloud Run sidecar, mri_deface, automated trigger
- Phase 3: Operations**
Routing rules, institution management, audit logging
- Phase 4: Advanced**
Burned in PHI OCR, QC automation, BIDS conversion

1. Uploader opens AEGIS in their browser (no install)
2. Drags DICOM folder into the upload portal (any modality)
3. Browser parses files and strips all PHI tags (PS3.15 Annex E)
4. Uploader reviews before/after anonymization preview
5. Confirms → encrypted upload to GCP (only de-identified data leaves site)

6. Server validates de-identification completeness
7. Head imaging → automated defacing; all other modalities → direct to clean state
8. Admin reviews quality in OHIF viewer, approves or rejects
9. Approved study is routed to destination project/institution

Key: Patient-identifying data **never leaves the hospital network**. The de-identification engine is **modality-agnostic** — it works identically whether the study is a brain MRI, a chest X-ray, or a cardiac ultrasound.

Supported Modalities

Category	Modalities
Cross-sectional	MRI, CT, PET, PET/CT, PET/MR
Projection	X-Ray (CR/DR), Mammography (FFDM/DBT), Fluoroscopy
Nuclear Medicine	SPECT, Planar Scintigraphy
Ultrasound	Echocardiography, Abdominal, Vascular, OB/GYN
Radiation Therapy	RT Structure, RT Plan, RT Dose, RT Image
Other DICOM	Secondary Capture, Structured Reports, Key Images, Presentation States

MVP focus: Brain MRI, PET, and CT — these exercise the full pipeline including automated defacing. All other modalities use the same upload and de-identification flow.

Technology Stack

Layer	Technology	Why
Backend	Go on Cloud Run	~10 MB container, 100ms cold starts, minimal CVE surface

Frontend	React + TypeScript	Modern, fast, PWA-capable
Database	Cloud SQL (PostgreSQL 15)	Relational data: users, projects, audit trail
DICOM Storage	GCP Healthcare API	Managed DICOMweb, built-in de-id validation
Analytics	BigQuery	DICOM metadata export, compliance reporting
AI/ML	Vertex AI	Burned-in PHI detection, image QC automation
Defacing	Python (mri_deface)	Proven facial feature removal, isolated sidecar
Viewer	OHIF Viewer	MIT-licensed, DICOMweb-native, all modalities
Infrastructure	Terraform	Reproducible, auditable IaC
Email	PSC→on-prem SMTP (internal) + SendGrid (external)	Dual-path, no PHI in notifications

Competitive Landscape

	AEGIS	XNAT	Flywheel	MIRC CTP
Hosting	GCP managed	Self-hosted	Commercial SaaS	On-prem
Client install	None (browser)	Desktop app	CLI tool	Java app
Modalities	All DICOM	Neuroimaging	All imaging	All imaging
Defacing	Automated	Manual/plugin	Built-in	None

Open source	Yes	Yes	No	Yes
DICOM store	Healthcare API	PostgreSQL + FS	MongoDB + S3	Filesystem

Differentiators:

1. Zero-install browser anonymization for any DICOM modality
2. Modality-agnostic — not locked to a single specialty
3. GCP-native with Healthcare API — managed DICOMweb, built-in de-id, BigQuery analytics
4. Automated defacing pipeline for head imaging
5. Vertex AI for burned-in PHI detection and image QC
6. Minimal attack surface (Go + distroless)

Phased Roadmap

Phase 1 — MVP (Brain MRI/PET/CT)

- GCP project + infrastructure (Terraform)
- Upload Portal with browser-based DICOM anonymization (all modalities)
- Go API with signed upload and Healthcare API ingest
- Admin Dashboard with OHIF viewer for QC
- Cloud SQL for app data, BigQuery for analytics

Phase 2 — Defacing + Notifications

- Automated mri_deface pipeline on Cloud Run (head imaging)
- Before/after comparison in admin dashboard
- Email notifications: SendGrid (external), SMTP/PSC (internal)

Phase 3 — Operations + Multi-Modality Expansion

- Multi-project routing engine for all modalities

- Institution onboarding and RBAC
- Configurable anonymization profiles per modality/project
- Audit reporting and compliance dashboards

Phase 4 — AI/ML

- Vertex AI burned-in PHI detection (all modalities)
- Automated image quality assessment
- Smart routing with modality/anatomy classification
- BIDS format conversion for research output

Cost Estimate (Dev/POC)

Resource	Monthly Cost
Cloud Run (API + dashboard)	~\$5-15 (free tier covers dev)
Cloud SQL (db-f1-micro)	~\$10
Healthcare API (DICOM store)	~\$0.50/GB stored
Cloud Storage (staging)	~\$0.02/GB
BigQuery	Free tier (1 TB queries/month)
Vertex AI	Pay per prediction (minimal during dev)
SendGrid	Free tier (100 emails/day)
Total (dev)	~\$20-40/month

Production costs scale with data volume. A typical multi-site research study with 1,000 imaging sessions (~500 GB) would cost approximately \$50-100/month in storage.



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Ready to Build

AEGIS enables secure, zero-install medical image sharing across all DICOM modalities — with automated anonymization, intelligent defacing, and AI-powered quality control — built on GCP with HIPAA compliance from day one.